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Berkeley, CA | [oliverhchang@berkeley.edu](mailto:oliverhchang@berkeley.edu) | (408) 203-9356

## EDUCATION

**University of California, Berkeley**

Class of 2028

*B.S. Mechanical Engineering | GPA: 3.91*

## SKILLS

**CAD/CAM:** Onshape, Solidworks, Fusion 360, AutoCAD, ANSYS Workbench, ANSYS Mechanical, ANSYS Spaceclaim

**Programming:** MATLAB, Python, Java, Arduino

**Manufacturing:** 3D Printing, BambuStudio, CNC, Fablight, Laser Cutting, Lathe Machines, Milling Machines, Waterjet

## EXPERIENCE

**Formula Electric at Berkeley**

Sept 2024 - Present

*Chassis Mechanical Engineer*

- Optimization of spaceframe stiffness by 56% from initial iterations using structural analysis with ANSYS, informing rule compliant chassis geometry and improved steering.
- Validated rule compliance and ensured 2+ Factor of Safety under crash, cornering, and braking scenarios
- Designed torsional stiffness testing jig and testing plan to validate simulation data and inform handling performance

**UC Berkeley's Space Technology and Rocketry (STAR)**

Sept 2024 - Present

*Airframe & Payload Engineer*

- Designed K-Class rocket layout on OpenRocket, conducted CFD simulations, fabricated airframe and recovery parts with a group of 8 all within a single semester
- Assembled, soldered, and programmed custom avionic and payload boards for altitude, pressure, and temperature data
- Led full development process of custom insulated Arduino-powered ice-cream maker payload, utilizing rapid prototyping and 3D printing to create a fully-functioning device over 3 iterations

**Fremont High Robotics FRC Team 3501: Firebots**

2020 - 2024

*VP of Mechanical Design*

- Led a team of 6 to design a competition robot for the 2024 FIRST Robotics Competition. The first iteration was designed within the first 2 weeks of build season, with an overhaul of the intake and flywheels after the first competition
- Coordinated with manufacturing to create industry-standard drawings & deliver high-quality parts
- Redesigned school workshop by designing and building 13 custom tables, 2 CNC Enclosures, with overhauled of tool and machine organization to streamline work processes
- Finished top 15% of all teams in the world

*Business Lead*

- Led a team to apply for 15+ grants, overhauled grant writing and budgeting systems from the ground-up
- Planned and hosted 35+ outreach events exposing 10,000+ attendees to FIRST opportunities
- Created 5 unique lesson plans to introduce STEAM to elementary school students, now taught monthly to 500+ students across 6 local elementary school after-school programs

## WORK EXPERIENCE

**Scouts BSA Summer Camp Staff**

June - July 2022, 2023, 2024

*Kitchen Staff, Aquatics Staff, Handicraft Director*

- Designed and taught custom Handicraft curriculum for Cub Scout utilizing leatherworking & woodworking skills
- Instructed over 400+ scouts on first aid, swimming, and lifesaving merit badges

## HONORS & AWARDS

*Dean's List Semifinalist, FIRST Robotics*

*Eagle Scout, Scouts BSA:* Drafted, financed, and built originally designed equipment shed for Sunnyvale Little League, working with local suppliers to reduce costs